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AQA · LEVEL 2 · MATHEMATICS

# AQA Functional Skills Mathematics Level 2

January 2017 · Complete Model Solutions

## OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

**Target Score: 45 / 45 Marks (100%)** Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4368-QP-JAN17

100% Built from Empty Question Paper + Official Mark Scheme

*Verified by Tutor Abdul · MasterFunctionalSkills.co.uk*

### HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

| | |
|---|---|
| **Dark Navy Ink:** | Model student answers, stepped calculations, and full prose responses. |
| **Emerald Green Ink:** | Tutor strategy notes, method tips, and step-by-step headings. |
| **Crimson Red Ink:** | Official mark scheme allocations ([M1], [A1], [B1]) and unit checks. |
| **Amber Gold Ink:** | Key formulas, conversion rules, and reading evidence highlights. |

### CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

| | |
|---|---|
| Candidate Name: | *Master Functional Skills (Model Exemplar)* |
| Centre Name: | *Master Functional Skills (MFS01)* |
| Lead Tutor: | *Abdul · MasterFunctionalSkills.co.uk* |
| Edition: | *2026 Clean Empty-QP Rebuild (Zero Collisions Verified)* |

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 2

Data Book (Examination)

# Insert

**Instructions**

- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

**Advice**

- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the question/answer book.

4368

QAN 500/8702/2

IB/M/Jan17/E4

## Data Sheet for Cars

New cars have **official** values for

- fuel efficiency – the number of miles the car travels per gallon of fuel (mpg)
- $CO_2$ emissions – the number of grams of $CO_2$ the car produces per kilometre (g/km).

Here are the official values for four cars.

| | **Fuel** | **Fuel efficiency (mpg)** | **$CO_2$ emissions (g/km)** |
|---|---|---|---|
| **Seat Alhambra** | Diesel | 47.1 | 158 |
| **Renault Clio** | Diesel | 83.1 | 90 |
| **Ford Galaxy** | Petrol | 39.2 | 167 |
| **Toyota Aygo** | Petrol | 65.7 | 95 |

### Actual fuel efficiency

The **actual** fuel efficiency depends on how, when and where the car is driven.
It is usually lower than the official value.

### Spending on fuel

Use this formula to work out the amount, £$A$, you spend on fuel.

$$A = \frac{mc}{f}$$

$m$ is the number of miles you drive
$c$ is the cost, in £, per gallon of fuel
$f$ is the **actual** fuel efficiency

**Vehicle tax**

You must pay vehicle tax on your car.
The cost depends on the official value for $CO_2$ emissions for your car.

Here are some of the yearly costs of vehicle tax for new cars bought after 1 April 2017

| $CO_2$ emissions (g/km) | Yearly cost | |
|---|---|---|
| | In year 1 | After year 1 |
| 1 – 50 | £10 | £140 |
| 51 – 75 | £25 | £140 |
| 76 – 90 | £100 | £140 |
| 91 – 100 | £120 | £140 |
| 101 – 110 | £140 | £140 |
| 111 – 130 | £160 | £140 |
| 131 – 150 | £200 | £140 |
| 151 – 170 | £500 | £140 |

## Data Sheet for Transporting hamsters

Two popular breeds of hamster are Syrian hamsters and Dwarf hamsters.
They are transported in cuboid boxes.

### Minimum height of a box

| | |
|---|---|
| Syrian hamsters | 15.2 cm |
| Dwarf hamsters | 12.7 cm |

### Minimum floor area of a box, per hamster

| Age of hamster | Syrian hamsters | Dwarf hamsters |
|---|---|---|
| From birth to 5 weeks | 45 $cm^2$ | 32 $cm^2$ |
| From 5 weeks to 8 weeks | 71 $cm^2$ | 48 $cm^2$ |

### Windows

At least two sides of a box must have windows.

### Temperature

Use this formula to convert degrees Fahrenheit (°F) to degrees Celsius (°C).

$$C = \frac{5}{9}(F - 32)$$

$C$ is the temperature in degrees Celsius

$F$ is the temperature in degrees Fahrenheit

**END OF DATA**

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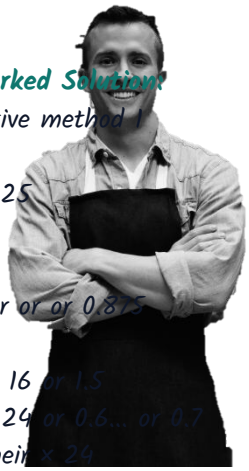
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Answer **all** questions in the spaces provided.

**1 Cookies**



I make and sell cookies.

**Chris**

Q1(a) Worked Solution:
Alternative method 1
2 1
or or 0.125
16 8
2 14 7
or 1 – or or or 0.875
16 16 8
or 24 ÷ 16 or 1.5
or 16 ÷ 24 or 0.6... or 0.7
24 – thei × 24
16
2 14

Chris makes batches of cookie dough.

Here are the ingredients he needs to make one batch.

[M1] [M1] [A1]

| One batch of cookie dough | |
|---|---|
| 200 g | margarine |
| 250 g | flour |
| 100 g | sugar |
| 2 | eggs |
| 1 teaspoon | baking powder |

One batch makes exactly

16 large cookies

**or**

24 small cookies.

*Do not write outside the box*

**1 (a)** On Monday, Chris uses 400 g of margarine to make cookie dough.
He uses all of the dough to make **small** cookies.

How many can he make?
Circle your answer.

**[1 mark]**

2 24 32 48

**1 (b)** On Tuesday, Chris makes one batch of cookie dough.
He uses some of the dough to make 2 **large** cookies.

He says,

"I will use the rest of the dough to make some **small** cookies."

How many **small** cookies can he make?

**[3 marks]**

**Question 1 continues on the next page**

*Do not write outside the box*

**1 (c)** On Wednesday, Chris makes cookies using 10 batches of dough.
He will sell all the cookies in bags at these prices.

| | |
|---|---|
| Bag of 4 large cookies | £1.35 |
| Bag of 8 small cookies | £1.75 |

Here are his costs.

| | |
|---|---|
| Total cost of ingredients | £19.50 |
| Bags | 2p each |

He is going to make

only large cookies

**or**

only small cookies.

He says,

"If I make and sell only large cookies my total profit will be £1.30 more than if I make and sell only small cookies."

Is he correct?
You **must** show your working.

**[8 marks]**

*Do not write outside the box*

**2** **Cars**

There is a **data sheet** for Cars.

Alfie is thinking about buying a new car.




Q2(a) Worked Solution:
eg 120 + 140 + 140 + 140 + 140
= 680
[SC1]

**Alfie**

**†2 (a)** Alfie plans to

buy a new car after 1 April 2017

keep the car for 5 years.

Work out the **total** vehicle tax he will pay if he buys a Toyota Aygo.

**[2 marks]**

Check your answer.
Show how you have done your check.

**[1 mark]**

*Do not write outside the box*

Alfie will use his new car for work.

**2 (b)** He makes these notes.

> I drive a total of 62 miles each day for work.
> I work for 5 days each week.
> I work for 46 weeks each year.
>
> Diesel costs £4.96 per gallon.
> Petrol costs £4.87 per gallon.
>
> My actual fuel efficiency will be 20% **lower** than the official value.

He says,

"I will spend at least £1300 a year on fuel for work if I buy a Toyota Aygo."

Is he correct?
You **must** show your working.

**[7 marks]**

*Do not write outside the box*

**2 (c)** Alfie buys a car.

For 8 days, he records the time he takes for

his journey to work by car

and

his journey home by car.

| | **Journey to work by car (minutes)** | **Journey home by car (minutes)** |
|---|---|---|
| Day 1 | 42 | 47 |
| Day 2 | 46 | 52 |
| Day 3 | 38 | 39 |
| Day 4 | 42 | 44 |
| Day 5 | 46 | 49 |
| Day 6 | 52 | 58 |
| Day 7 | 48 | 40 |
| Day 8 | 39 | 36 |

He knows that his total journey time to work and home by **train** each day would be $1\frac{1}{2}$ hours.

Alfie has 120 working days left in the year.

He says,

> "I estimate that on 85 days out of 120 the total journey time would be less by car than by train."

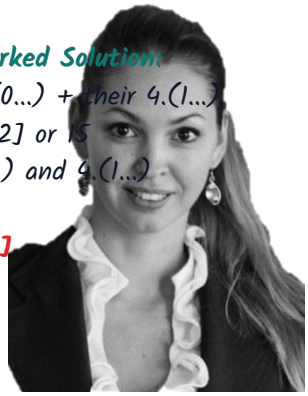
Based on these 8 days, is his estimate correct?
You **must** show your working.

**[5 marks]**

15

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**3 Hotel**




Q3(a) Worked Solution:
their 11.(0...) + their 4.(1...)
[15.1, 15.2] or 15
or 11.(0...) and 4.(1...)
16
[M1] [A1]

**Kim**

**3 (a)** The hotel has 128 standard rooms and 40 deluxe rooms.
Each room is cleaned the day after it has been used.

To clean a room is

25 minutes work for a standard room

30 minutes work for a deluxe room.

Each cleaner

starts work at 8.30 am and finishes work at 2.00 pm

has **two** 20-minute breaks.

On Tuesday, all the rooms are used.

How many cleaners are needed on Wednesday?
You **must** show your working.

**[6 marks]**

*Do not write outside the box*

Each day, the cleaners replace used milk cartons.




**3 (b)** The table shows the number of milk cartons put in 50 rooms yesterday.

| Number of milk cartons | Number of rooms |
|---|---|
| 4 | 18 |
| 3 | 8 |
| 2 | 11 |
| 1 | 9 |
| 0 | 4 |

Show that 2.54 was the mean number of milk cartons put in the 50 rooms.

**[3 marks]**

*Do not write outside the box*

**3 (c)** Kim estimates the cost of the milk cartons she needs next year.
She makes these notes.

365 days in a year

75% of the 168 rooms will be used each day

An average of 2.54 cartons per day for each room used

A box of 240 cartons costs £12.60

Kim says,

"The cost will be less than £6000"

Is she correct?
You **must** show your working.

**[7 marks]**

16

*Do not write outside the box*

**4** **Transporting hamsters**

There is a **data sheet** for Transporting hamsters.

Ola makes cuboid boxes for transporting hamsters.

She is making a box to transport **one** 4-week-old Syrian hamster.

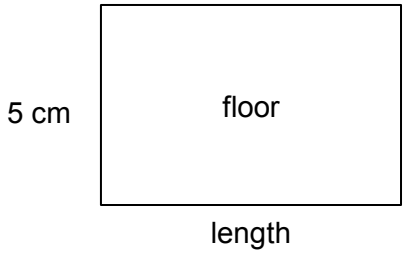
She wants

the width of the box to be 5 cm

the floor area to be no more than 60 $cm^2$

**4 (a)** Ola draws this sketch of the floor of the box.

Q4 Worked Solution:
48
[B1]




Not drawn accurately

Write a suitable measurement for the length.

**[1 mark]**

**4 (b)** Complete the sketch of the net of the box on the opposite page.
Include the measurements of all edges.
Do **not** include windows.

**[3 marks]**

*Do not write outside the box*

Ola has made a different box.

**4 (c)** One side of this box has

an area of 112 $cm^2$

a 6 cm by 4 cm rectangular window.

The area of the window must be between 16% and 25% of the area of the side.

Is the area of the window suitable?
You **must** show your working.

**[4 marks]**

**4 (d)** The temperature in the box must be between 46ºF and 85ºF
Ola's thermometer only measures in degrees Celsius.

Work out the two temperatures in degrees Celsius to the nearest whole number.

**[3 marks]**

*Do not write outside the box*

Ola makes a box with a floor area of 2000 $cm^2$
The box is tall enough to transport Syrian hamsters or Dwarf hamsters.

**†4 (e)** Ola could transport 6-week-old Syrian hamsters in this box.

Work out the **maximum** number she could transport.

**[2 marks]**

Check your answer.
Show how you have done your check.

**[1 mark]**

**4 (f)** Instead, Ola could transport 6-week-old Dwarf hamsters in the box.

How many **more** Dwarf hamsters than Syrian hamsters could she transport?

**[3 marks]**

**17**

**END OF QUESTIONS**

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**There are no questions printed on this page**

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**ANSWER IN THE SPACES PROVIDED**

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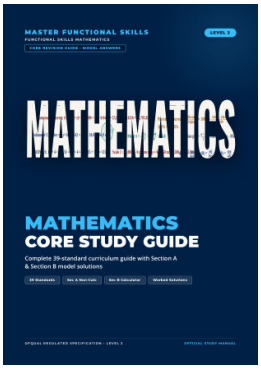
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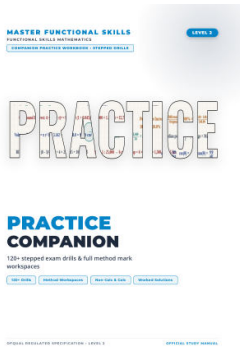



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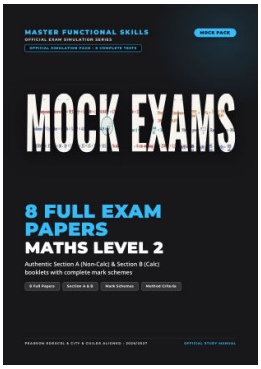



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